

Program : Diploma in Civil and Environmental Engineering	
Course Code: 6351B	Course Title: Solid Waste Management and Air Pollution control
Semester : 6	Credits: 4
Course Category: Programme Elective Course	
Periods per week: 4(L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To enable students to familiarize with various functional elements involved in Solid Waste management.
- To enable students to discuss various methods for solid waste disposal.
- To enable students to identify impacts of air pollution and discuss various methods for air pollution control.

Course Prerequisites:

Topic	Course code	Course name	Semester
Solid Waste management, Air pollution	2001	Environmental Science	2
Sustainable development	4353	Sustainable engineering	4

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Classify solid waste based on source and characteristics and identify steps in waste collection, storage and transportation.	14	Understanding
CO2	Discuss various methods for disposal of solid waste based on waste characteristics.	15	Understanding
CO3	Identify sources, impacts and sampling and measurement methods of air pollutants.	15	Understanding
CO4	Discuss various methods for air pollution control.	14	Understanding

	Tests	2	
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CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2		3		
CO2	3	2	2		3		
CO3	3	2	2		3		
CO4	3	2	2		3		

3-Strongly mapped, 2-Moderately mapped,1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Classify solid waste based on source and characteristics and identify steps in waste collection, storage and transportation.		
M1.01	Classify solid waste based on source.	1	Understanding
M1.02	Describe characteristics of each category of waste.	5	Understanding
M1.03	Identify purpose and functional elements of solid waste management.	3	Understanding
M1.04	Describe waste collection, storage and transportation system	5	Understanding
	Total	14	

Contents:

Introduction to the course

Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes.

Hazardous waste- Identification and classification of hazardous waste.

Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016

Waste Collection, Storage and Transport: Collection

components, storage containers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system.

Processing techniques- Mechanical volume and size reduction, chemical volume reduction, component separation.

CO2	Discuss various methods for disposal of solid waste based on waste characteristics.		
M2.01	Discuss bio chemical methods for disposal of solid wastes.	6	Understanding
M2.02	Discuss thermo chemical methods for disposal of solid wastes.	6	Understanding
M2.03	Discuss treatment methods for hazardous wastes	3	Understanding
	Series 1	1	
	Total	16	

Contents

Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages.

Composting- Stages in aerobic composting, types of composting-Indore and Bangalore process

Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods

Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages.

Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill.

CO3	Identify sources, impacts and sampling and measurement methods of air pollutants.		
M3.01	Identify sources, characteristics of modern air pollutants.	6	Understanding
M3.02	Identify effects of air pollutants of environment and health.	4	Understanding
M3.03	Discuss sources and impacts of indoor air pollution.	2	Understanding
M3.04	Explain air sampling and pollution measurement methods.	3	Understanding

	Total	15	
Contents: Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used.			
CO4	Discuss various methods for air pollution control.		
M4.01	Identify air pollution indices available in India for air quality monitoring.	2	Understanding
M4.02	Discuss various control methods for particulate pollutants.	5	Understanding
M4.03	Discuss various control methods for gaseous pollutants.	5	Understanding
M4.04	Discuss various biological air pollution control techniques.	2	Understanding
	Series 2	1	
	Total	15	
Contents: Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers.			

Text / Reference:

T/R	Book Title/Author
T1	Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi , Latest Edition
T2	C. S Rao, “Environmental Pollution Control Engineering” , New Age International Publishers, 2006
T3	R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008
R1	M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi 2017
R2	CPHEEO. Manual on Municipal solid waste management. Central public Health and Environmental Engineering Organization, Government of India, New Delhi., 2016
R3	Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , McGraw Hill,New Delhi, 1985.
R4	Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, 2023

Online resources:

Sl.No.	Website Link
1	https://cpcb.nic.in/
2	https://nptel.ac.in/